Two Counterexamples to Conjectures in *Dynamical Sampling: A Survey*

Literature-status solution packet

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Provenance notice. This packet records later literature answers; it does not claim a new result. Conjecture 1 was explicitly refuted in arXiv:2605.29671, and Conjecture 3 was explicitly refuted in arXiv:2606.20848. The prime example below is a concrete specialization of the argument in Example 3.4 of arXiv:2605.29671.

1 Source questions and outcome

The source is A. Aldroubi, C. Cabrelli, I. Krishtal, and U. Molter, *Dynamical Sampling: A Survey*, arXiv:2511.10769v3. It asks, in Conjecture 1 (typeset page 9), whether the following implication holds. If $\{D^\ell b : \ell = 0, 1, \ldots\}$ is a Carleson frame with $\sigma(D) \subset (0, 1)$, and $\Lambda \subset (0, \infty)$ obeys

$$\sum_{\lambda \in \Lambda} \frac{1}{\lambda} = \infty,$$

then $\{D^\lambda b : \lambda \in \Lambda\}$ should be a frame whenever it is Bessel.

Conjecture 3 (typeset page 19) says that if A is bounded and normal on an infinite-dimensional Hilbert space and $G \subset H$, then the family

$$\left\{ \frac{A^n g}{\|A^n g\|} : g \in G,\ n \geq 0 \right\}$$

cannot be a frame.

Both conjectures are false. The first fails already for a positive diagonal operator and the prime sampling times. The second fails even for a singleton $G = \{g\}$ and a bounded invertible normal operator on $\ell^2(\mathbb{N})$.

2 A concrete counterexample to Conjecture 1

Let $H = \ell^2(\mathbb{N})$, indexed by $j \geq 1$, and put

$$r_j = 1 - 2^{-j}, \qquad De_j = r_j e_j, \qquad b = \sum_{j \geq 1} \sqrt{1 - r_j^2}\, e_j.$$

The vector b belongs to ℓ^2, since $\sum_j (1 - r_j^2) < \infty$. Let $\mathcal{P}$ denote the set of primes.

Theorem 1. *The integer orbit $\{D^n b : n \geq 0\}$ is a Carleson frame, while the prime suborbit $\{D^p b : p \in \mathcal{P}\}$ is Bessel and complete but is not a frame. Moreover, $\sum_{p \in \mathcal{P}} p^{-1} = \infty$.*

Proof. We first verify the only non-immediate part of the Carleson-frame criterion. For $k = j + m$, direct calculation gives

$$\frac{|r_j - r_k|}{|1 - r_j r_k|} = \frac{1 - 2^{-m}}{1 + 2^{-m} - 2^{-j-m}} \geq \frac{1 - 2^{-m}}{1 + 2^{-m}} =: a_m.$$

The same lower bound holds when $k = j - m$. Because

$$\sum_{m\geq 1}(1 - a_m) = \sum_{m\geq 1} \frac{2^{1-m}}{1 + 2^{-m}} < \infty, \qquad P := \prod_{m\geq 1} a_m > 0,$$

we obtain, uniformly in j,

$$\prod_{k\neq j} \frac{|r_j - r_k|}{|1 - r_j r_k|} \geq P^2 > 0.$$

Thus (r_j) is a Carleson sequence. We also have $0 < r_j < 1$, $r_j \to 1$, and the coefficient of e_j in b is exactly $\sqrt{1 - r_j^2}$. The diagonal Carleson-frame characterization (Theorem 2.9 of the survey) now shows that $\{D^n b\}_{n\geq 0}$ is a frame.

Every subfamily of a Bessel family is Bessel, so the prime suborbit is Bessel. On the unit coordinate vector e_j, its analysis energy is

$$E_j := \sum_{p\in\mathcal{P}} |\langle e_j, D^p b\rangle|^2 = (1 - r_j^2) \sum_{p\in\mathcal{P}} r_j^{2p}.$$

Set $s_j = r_j^2$. The primes have natural density zero, so the indicator sequence $c_n = \mathbf{1}_{\{n\in\mathcal{P}\}}$ has Cesàro mean zero. Abel's theorem for Cesàro means yields

$$(1 - s) \sum_{n\geq 1} c_n s^n \longrightarrow 0 \qquad (s \uparrow 1).$$

Consequently $E_j \to 0$, which contradicts any positive lower frame bound. Hence the prime suborbit is not a frame.

Euler's theorem gives $\sum_{p\in\mathcal{P}} 1/p = \infty$, exactly the Müntz–Szász condition in Conjecture 1. The same condition guarantees completeness here (as recalled in both later supporting papers), so this example cleanly separates completeness from the stable lower frame bound. □

This is the prime case in Example 3.4 of E. Gallardo-Gutiérrez and J. Partington. Their paper, arXiv:2605.29671, explicitly says it gives counterexamples to the survey conjecture and proves the same vanishing of the lower-bound test. The elementary Carleson calculation above merely makes one fully concrete choice of the starting frame.

3 The later density classification

The subsequent paper I. Krishtal and B. Miller, arXiv:2607.18491, provides a sharper structural explanation. Its Theorem 1.3 says that if $\{J^k g\}_{k\geq 0}$ is a positive-spectrum block-diagonal Carleson frame and the strictly increasing sampling sequence Λ has natural density

$$L = \lim_{x\to\infty} \frac{N_\Lambda(x)}{x},$$

then $\{J^\lambda g\}_{\lambda\in\Lambda}$ is a frame if and only if $0 < L < \infty$. Ordinary diagonal Carleson frames are the rank-one-block case. Since the primes have density $L = 0$, this theorem subsumes the counterexample above. The same paper explicitly notes that the Müntz condition is necessary but insufficient for a lower frame bound.

Thus the obstruction is not failure of the Müntz completeness mechanism; it is asymptotically too sparse a sampling rate for stable reconstruction.

4 The explicit counterexample to Conjecture 3

I. Krishtal and G. Pfander, *The Normalized Orbit of a Bounded Normal Operator Can Be a Frame*, arXiv:2606.20848, names Conjecture 3 of the survey in its abstract and states that it is false. Its main theorem is:

> There exist a bounded invertible normal operator T on $\ell^2(\mathbb{N})$ and $g \in \ell^2(\mathbb{N})$ such that $\{T^k g/\|T^k g\| : k = 0, 1, 2, \ldots\}$ is a frame.

Taking $G = \{g\}$ gives exactly a counterexample to the universally quantified statement in Conjecture 3. The construction uses a diagonal normal operator on rapidly growing finite blocks. During a long interval of times, the normalized orbit is concentrated in one block and follows a Fourier basis there. A selected subsequence is a small perturbation of an orthonormal basis, yielding a lower frame bound, while a Schur-test estimate provides the upper Bessel bound for the entire normalized orbit. These are the two logically necessary parts of the frame claim; their detailed proof is contained in the supporting paper included with this packet.

5 Status, scope, and verification

Source item	Status	Primary evidence
Conjecture 1	False	arXiv:2605.29671, Example 3.4; independently subsumed by arXiv:2607.18491, Theorem 1.3.
Conjecture 3	False	arXiv:2606.20848, abstract and Theorem 1.

This packet makes no claim about the survey's other open directions: frame indices, multi-orbits, time discretization, nonlinear recovery, or system identification. It also makes no priority or novelty claim for the displayed prime construction. Its purpose is clerical but mathematically complete: identify two exact source conjectures, connect each to a direct later answer, and expose the short lower-frame obstruction for Conjecture 1.

The following checks were performed against the bundled primary PDFs:

1. exact source statements: Conjectures 1 and 3 in arXiv:2511.10769v3;

2. explicit prime example and lower-bound failure: Example 3.4 of arXiv:2605.29671v1;

3. explicit normalized-orbit refutation: abstract and Theorem 1 of arXiv:2606.20848v1;

4. density-zero classification: Theorem 1.3 and Section 5 of arXiv:2607.18491v1.

6 References

1. A. Aldroubi, C. Cabrelli, I. Krishtal, and U. Molter, *Dynamical Sampling: A Survey*, arXiv:2511.10769v3 (2026).

2. E. A. Gallardo-Gutiérrez and J. R. Partington, *Frame Constructions Associated with Operator Orbits*, arXiv:2605.29671v1 (2026).

3. I. A. Krishtal and G. E. Pfander, *The Normalized Orbit of a Bounded Normal Operator Can Be a Frame*, arXiv:2606.20848v1 (2026).

4. I. Krishtal and B. Miller, *Block Diagonal Carleson Frames*, arXiv:2607.18491v1 (2026).